(Please write your Enrollment Number) Enrollment No. 01001042022

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: B.Tech. MAE/DMAM, Semester: III
(November-December, 2023)

Subject Code: BAS 205 | **Subject: Numerical Techniques for Engineers**
Time : 3 Hours | **Maximum Marks :60**
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8 = 20)**	
	(a) How many digits are to be taken in computing $\sqrt{20}$ so that the error does not exceed 0.1% ?		
	(b) Find the absolute error and relative error for the sum, $S = \sqrt{3} + \sqrt{5} + \sqrt{7}$ to four significant digits.		
	(c) Write the iterative formula of SOR Method for the following system of equations $a_{11}x + a_{12}y + a_{13}z = b_1,$ $a_{21}x + a_{22}y + a_{23}z = b_2,$ $a_{31}x + a_{32}y + a_{33}z = b_3$.		
	(d) Find the condition number for the following system of equations, also verify ill conditioned or well-conditioned? $x + y = 1, 2x + 1.01y = 2.01$		
	(e) Show that $\Delta = E\nabla = \nabla E = \delta E^{-1/2}$		
	(f) Write Newton-Cote's quadrature formula in terms of y_0.		
	(g) Explain IVP (Initial value problem) with example.		
	(h) *Write formula for* (a) *Trapezoidal rule* (b) *Modified Euler's formula*		
	UNIT-I		
Q2	(i) Find a root in [2, 3] of the equation $x^3 - 4x - 9 = 0$, using the bisection method correct to four decimal places. (ii) Find the rate of convergence of Newton-Raphson's method.	**(10)**	
Q3	(i) Find the smallest positive root in [0, 1] correct to three decimal places of the equation $1 - x + \frac{x^2}{(2!)^2} - \frac{x^3}{(3!)^2} + \frac{x^4}{(4!)^2} - \frac{x^5}{(5!)^2} \ldots\ldots = 0$ (ii) Using secant method find least positive real root of the equation $xe^x = 1$ correct to four decimal places.	**(10)**	
	UNIT-II		
Q4	Solve by LU decomposition (Crout's method with $u_{ii} = 1$ in Upper triangular Matrix U) $2x + y + 4z = 12,$ $8x - 3y + 2z = 20,$ $4x + 11y - z = 33$.	**(10)**	

Q5 By Rayleigh's power method find the largest eigen value and corresponding eigen vector. Take $[1 \quad 0 \quad 0]^T$ as initial eigen vector. **(10)**

$$A=\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

UNIT-III

Q6 Write formula for Simpson's 1/3'rd and Simpson's 3/8'Th rules and also evaluate $\int_0^1 \frac{dx}{1+x^2}$ by both the methods for n= 8 and n=9 respectively. **(10)**

Q7 **(10)**

(i) Apply Lagrange's interpolation formula and find the missing term f(3) = 19

x	0	1	2
f(x)	1	3	9

(ii) Determine the polynomial f(x) by Newton's divided difference interpolation formula

x	-4	-1	0
f(x)	1245	33	5

$5+66x+94x^2$

UNIT-IV

Q8 Evaluate $y(1.1)$ and $y(1.2)$ by Taylor's series method for the IVP $\frac{dy}{dx} = \log(xy)$, $y(1) = 2$. **(10)**

2.085
2.175

Q9 Solve the IVP for y(0.4) with h= 0.2 by Runge -Kutta method of 4'th order **(10)**

$$y' = -2xy^2, y(0) = 1.$$